

Matrix inequalities and norms

Dimension-independent Hlawka constants for Schatten norms

Difficulty: challenging **Importance:** interesting to the community**Status:** Partially resolved

Rating rationale: Sharp cancellation constants for entire Schatten classes require new control across the Schatten scale, a challenging norm-geometry question; dimension-independent control would inform perturbation estimates across the community.

Partial result — 2026-09-11

Partial result by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review: PASS for the stated partial result.**

The trace-norm endpoint has $C_1 = \infty$: a real 2×3 rank-one family has defect ratio asymptotic to $2/(3t)$ as $t \downarrow 0$. Zero-row padding gives the same failure on real 3×3 matrices.

Still open: The remaining finite Schatten exponents $1 < p < \infty$ are undetermined by this result; the known value $C_2 = 1$ is unchanged. The ratings apply to this surviving question.

Primary manuscript: [complete proof PDF](#), [standalone TeX](#), Theorem 1.1 and its proof; [authorship and scope](#). The [independent review](#) checks the full original argument and records its hash. The draft was AI-assisted; this is independent agent verification, not external human peer review or formal certification. [Submission record](#).

Problem statement

For each $1 \leq p < \infty$, determine the smallest $C_p \in [0, \infty]$ such that for every pair of positive integers N, M and every $X, Y, Z \in \mathbb{C}^{N \times M}$,

$$\|X\|_p + \|Y\|_p + \|Z\|_p - \|X + Y + Z\|_p \leq C_p (D_p(X, Y) + D_p(X, Z) + D_p(Y, Z)),$$

where $D_p(U, V) = \|U\|_p + \|V\|_p - \|U + V\|_p$ and $\|U\|_p = (\sum_j s_j(U)^p)^{1/p}$. The value $C_p = \infty$ means that no finite constant works uniformly over dimensions. The inequality must also hold when the parenthesized expression vanishes; a ratio definition must not discard such triples.

This asks whether the cancellation loss in a three-term matrix sum can be controlled sharply by the cancellation losses of its pairs. Such estimates concern the geometry of the matrix norms used in perturbation bounds and low-rank approximation.

Literature update — 2026-10-06

Ezzeri Esa’s September 2026 *Dimension-independent Hlawka constants for Schatten norms* proof release states that a finite dimension-independent constant exists for every $1 < p < \infty$. The pinned source declaration `PalomarHlawkaSchatten.dimension_independent_hlawka_constant_for_schatten_norms` uses the Schatten norm defined from singular values and quantifies over all finite-dimensional complex domain and codomain spaces. Its inequality includes triples with zero total pair defect, so its stated scope matches the finiteness part of this problem for rectangular complex matrices ([release README](#), [existence solution](#)). The author does not claim to determine the sharp constants for general operators.

The September 25, 2026 [Palomar record, version 1](#), lists three verified declarations in the *Construction-Diagonal* namespace within `PalomarHlawkaSchatten`:

- `diagonal_hlawka_bound`;
- `diagonal_hlawka_sharp`;
- `cyclic_maximum_attained`.

These concern an explicit sharp constant for complex diagonal matrices when $p \geq 256$, expressed as a one-variable maximum over $[1/2, 2]$ and attained by a three-coordinate cyclic construction. The bound is uniform in dimension, with sharpness in each dimension at least three ([pinned diagonal solution, registry comparator](#)). The registry’s listed declarations do not include the general-matrix existence theorem; its diagonal verification record must be distinguished from the author’s broader existence proof release.

Scope and remaining gap: The existence theorem and the sharp diagonal subcase are partial progress toward the original target. Determining the least constants for general complex matrices, beyond already settled cases such as $C_2 = 1$, remains open. The status remains **Partially resolved**. This update inspected the pinned source and registry record; it did not run Lean or independently audit the complete existence proof, and it does not assign full-target formal verification.

References

1. K. M. R. Audenaert and F. Kittaneh, *Problems and Conjectures in Matrix and Operator Inequalities*, arXiv:1201.5232v3 (2012), §8.2, Problem 7, equation (48), and the following discussion. [Full text](#).
2. Audenaert and Kittaneh, published version in *Études opératorielles*, Banach Center Publications 112 (2017), §8.2, Problem 6, equation (48), p. 25. [Publisher PDF](#).
3. E. Esa, *Dimension-independent Hlawka constants for Schatten norms*, September 2026 proof release, pinned at commit `1829591140522193a81ae721330e4d775c81123e`; general existence and diagonal construction are separate declarations. [Source](#). The diagonal construction is recorded as `PALOMAR-2026-09-25-000006`, version 1. [Registry record](#).

Status check (2026-09-10): the source gives $C_2 = 1$ and demonstrates nonexistence of a finite constant for the operator norm, while posing the Schatten constants generally. Current source records and searches for “Hlawka Schatten constant”, “Hlawka operators Audenaert Kittaneh”, and 2025/2026 found no determination of all C_p . Later determinantal Hlawka inequalities concern a different inequality and do not answer this norm-deficit question. The search is bounded; individual p -cases should be rechecked before treating them as new results.